

Logic :-

↳ Systematic method of reasoning that helps us to analyze statements & deductions based on their truth or falsity.

eg. Today is Sunday. $\rightarrow$ simple logical statement.

$\forall n \geq 0, 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \checkmark \rightarrow$ more complex logical statement.

$\hookrightarrow$ T/F

$$n=1, 1 = \frac{1(1+1)}{2} = 1$$

$$n=k, 1+2+\dots+k = \frac{k(k+1)}{2} \quad \longleftarrow \textcircled{1}$$

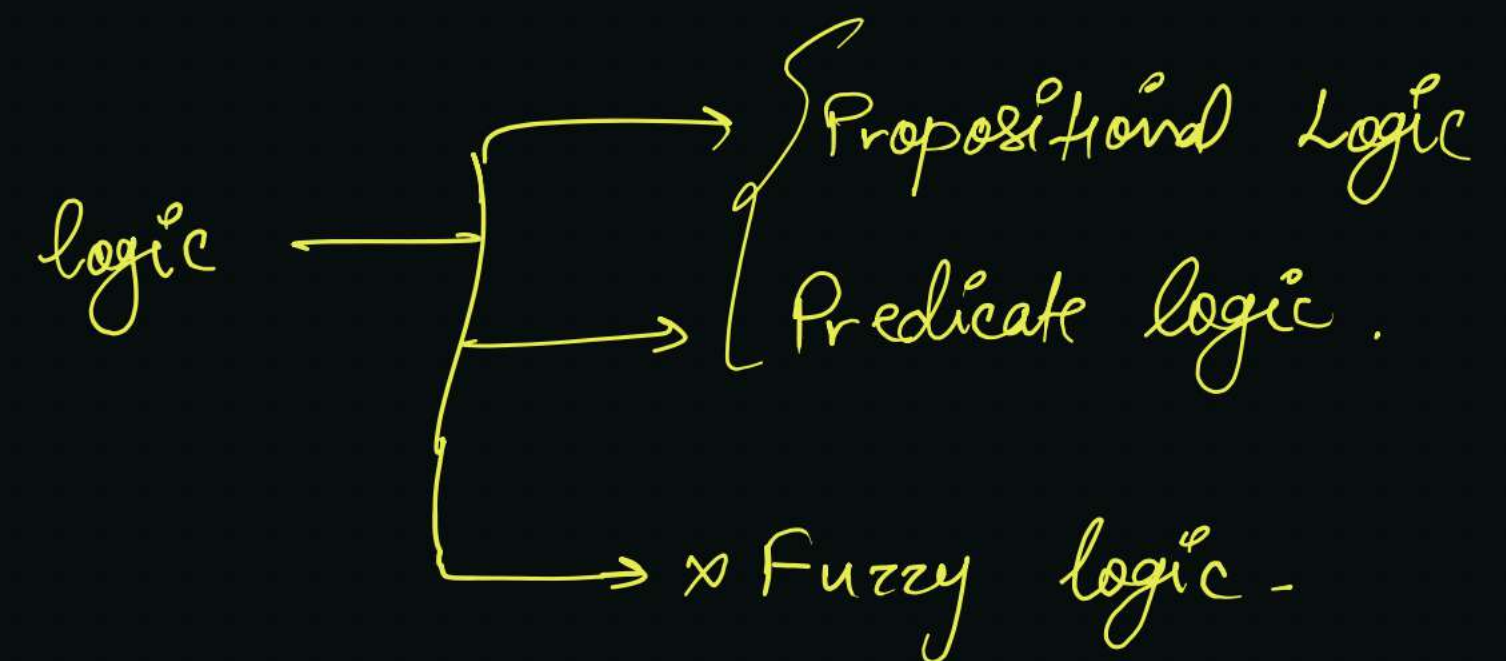
$$n=k+1, \underline{1+2+\dots+k} + k+1 = \frac{k(k+1)}{2} + (k+1)$$

$$= \frac{(k+1)(k+2)}{2} = \frac{n(n+1)}{2}$$

$n \equiv k+1$

Why Study Logic ?

- ↳ In maths → proofs, set theory, etc.
- ↳ In CS → programming (if, else, AND, ...)
digital circuits (logical gates)
DBMS & SQL (logical queries).
- ↳ In AI → knowledge representation & reasoning.
Inference.
Rule based models,
Fuzzy logic → multi-valued logic → approximate reasoning.
- ↳ In our day to day interactions → everyday reasoning to ensure clarity.
- ↳ Its part of GATE Q5.



Propositional logic :-

- ↳ These are propositions (simple declarative statements).
- ↳ That are either true or false but not both.

eg. $1+1 = 2$ ✓ (T)

$1+3 = 7$ ✓ (F)

What is the time? ✗

New Delhi is the capital of India. ✓ (T)

$5 > 10$ ✓ (F)

$x+1 = 3$ ✗ (neither T nor F). $\rightarrow$ (It depends on x)

Close the door. ✗ (is a command).

Note :- If you see a command, question or an ambiguous statement, all these are non-propositional in nature.

- ↳ We use letters to denote propositional variables (sentential variables)
 $p, q, r, a, b, c, \dots \rightarrow$ most simpler propositions $\rightarrow$ atomic propositions

Logical Connectives :- These symbols connect propositions to form compound statements.

1. Negation (NOT) :-

↳ Let p be the proposition, negation of p , $\neg p$ or $\bar{p}$, is "It is not the case that p ." or "not p ".

eg. p : My PC runs linux.

$\bar{p}$: It is not the case that my PC runs linux.
or

My PC does not run linux.

Syntax $\rightarrow \neg p$ or $\bar{p}$

Semantics $\rightarrow$ meaning of negation (Truth table).

↳ Truth table :-

p	$\neg p$	Syntax
T	F	
F	T	Semantics

2. Conjunction (AND) :-

↳ Let p & q be propositions.

The conjunction of p & q , denoted by $p \wedge q$, is the proposition " p and q ".

eg. p: I went to store.

9 : I bought some apples.

$p \wedge q$: I went to store and bought some apples;
 p q

↳ Truth table :-

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

3. Disjunction (OR) : (Inclusive OR)

↳ Let p & q be propositions.

Disjunction of p & q , denoted by $p \vee q$, is the proposition
" p or q " $\rightarrow$ at least one is $T \rightarrow T$ else F .

eg. p : You can go to party.

q : You can stay home & relax.

$p \vee q$: You can go to party or can stay home & relax.
 p q

↳ Truth Table :-

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

(Inclusive OR).

4. Exclusive OR :- (XOR).

↳ Let p & q be propositions.

Exclusive OR of p & q , denoted by $p \oplus q$, is the proposition that is T when exactly one of p & q is T otherwise its F.

Q. p : A student can have salad with dinner.
 q : A student can have soup with dinner.

$$p \oplus q = ?$$

Sol. $p \oplus q$: A student can have salad or soup, but not both,
in the dinner.

↳ Truth table :-

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

5. Conditional Statement (If then) :- (Implication).

↳ Let p & q be propositions.

The conditional statement, $p \rightarrow q$, is the proposition "if p then q " or " p implies q ".

$(p) \rightarrow (q)$ will only be F, when $p: T$ & $q: F$.

↳ conclusion or consequence.

↳ hypothesis or premise.

eg. p : It rains

q : We will stay indoors.

$p \rightarrow q$: If it rains, then we will stay indoors.

↳ Different ways to express conditional statements :-

1. "if p , then q " $\rightarrow$ " p implies q ". "implies, only if, if then & if p, q ."

2. "if p, q " $\rightarrow$ " p only if q ".

3. " p is sufficient for q " $\rightarrow$ "a sufficient condition for q is p " "A sufficient condition for staying indoors is it rains."

4. " q if p " $\rightarrow$ " q whenever p ".

5. " q when p " $\rightarrow$ " q is necessary for p ".

6. "a necessary condition for p is q " $\rightarrow$ " q follows from p ".

7. " q unless $\neg p$ " $\rightarrow$ " q provided that p ".

"we will stay indoors provided it rains."

"A necessary condition for it to rain is staying indoors."

"staying indoors follows from rains."

↳ Truth table :-

p	q	$p \rightarrow q$
T	T	T
<u>T</u>	F	<u>F</u>
F	T	T
F	F	T

eg. If I am elected, then I will lower the tax.
p q.

(H.W.) Construct truth tables of these compound propositions:

a) $p \rightarrow (\neg q)$

b) $(p \vee q) \rightarrow (p \wedge q)$

c) $(p \rightarrow q) \rightarrow (q \rightarrow p)$

d) $p \oplus (p \vee q)$

e) $\neg p \rightarrow (\neg q \vee r)$